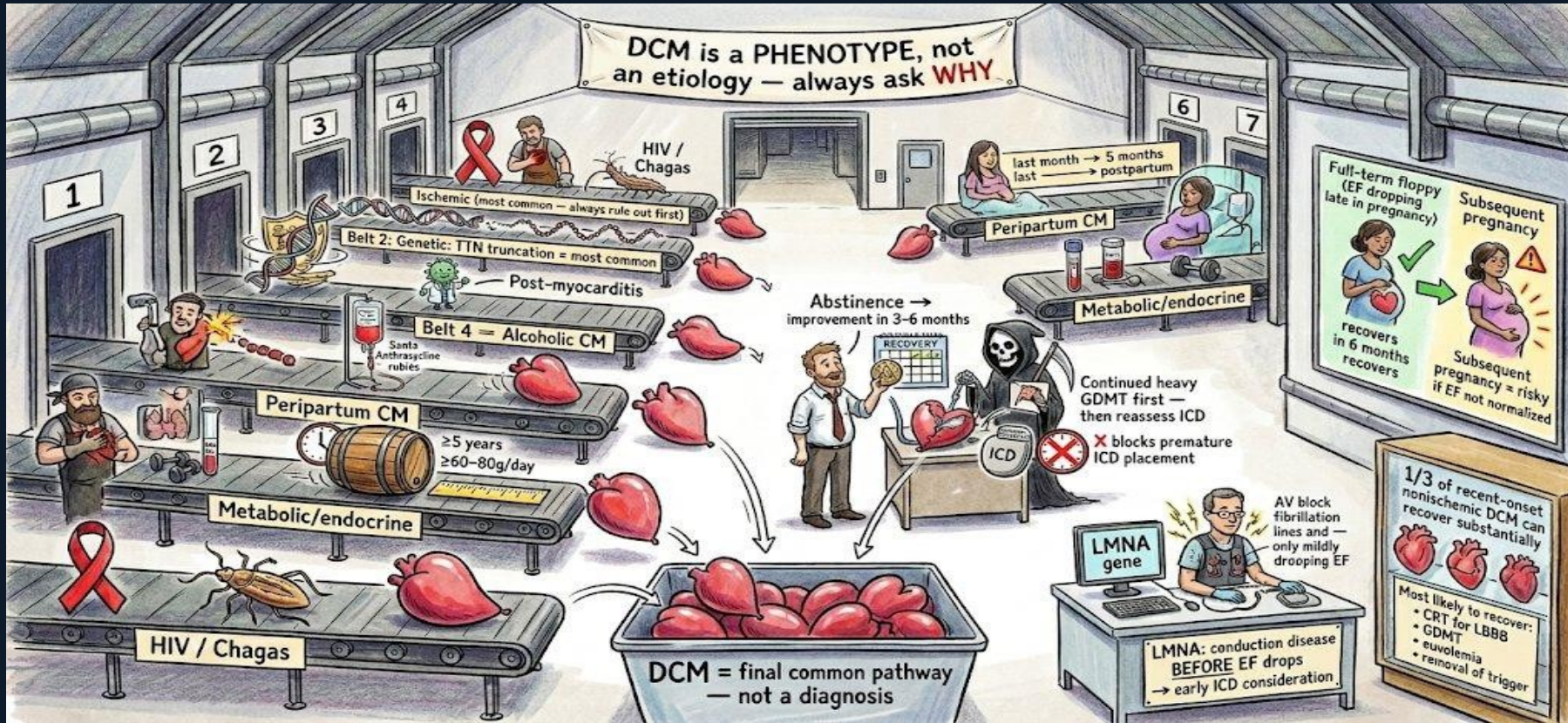
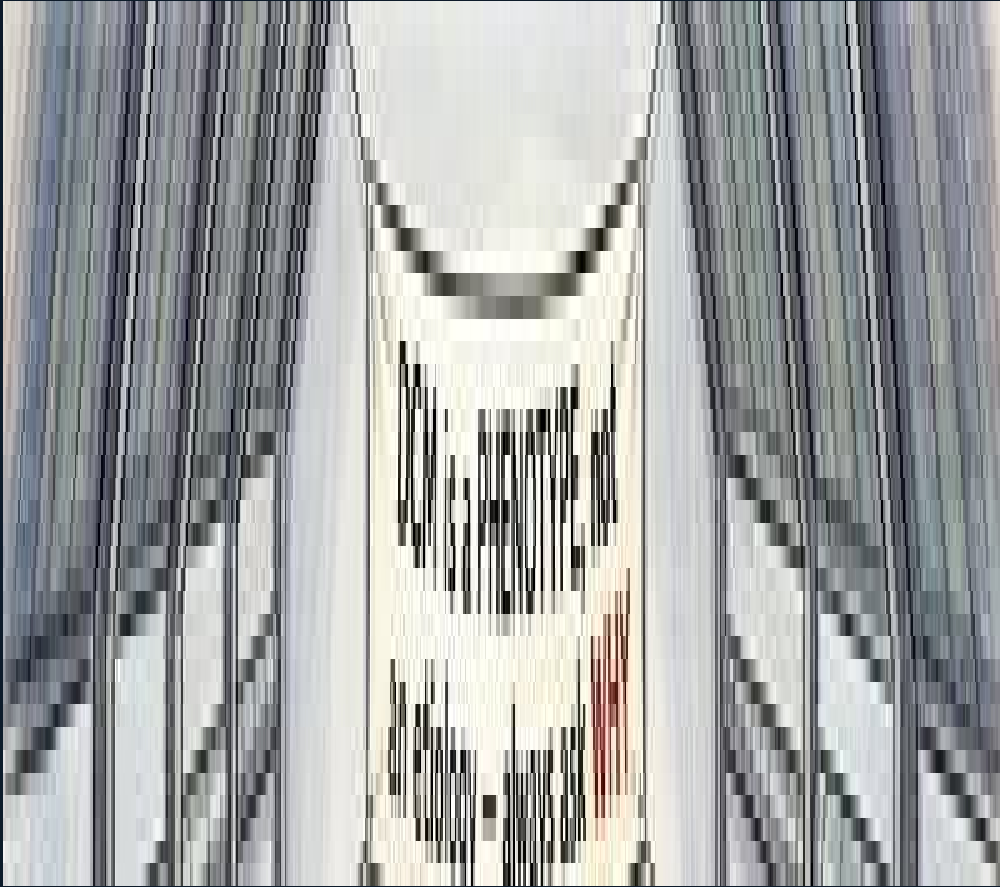


# Scene E — DCM: Causes & Alcoholic CM



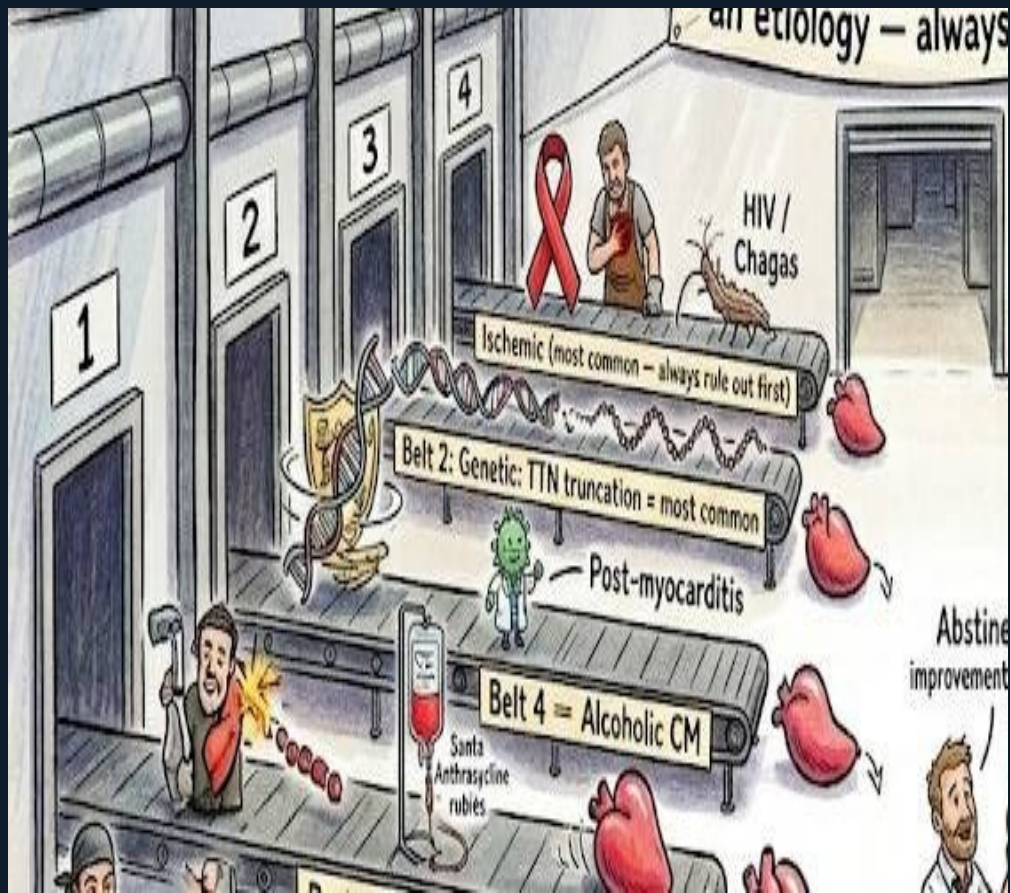
Industrial Factory · 4 board questions



## DCM is a PHENOTYPE, Not a Diagnosis

The factory banner above the entrance reads: 'DCM is a PHENOTYPE, not an etiology — always ask WHY.' This is the mandatory first principle. Dilated cardiomyopathy is not a diagnosis — it is a final common pathway that many different diseases produce. Calling something 'idiopathic DCM' is only permissible after excluding all reversible and heritable causes. The board tests this with: 'A patient has newly diagnosed DCM — what is the most important next step?' The answer is always a thorough etiologic workup — beginning with ruling out ischemia — not immediate GDMT or ICD placement.

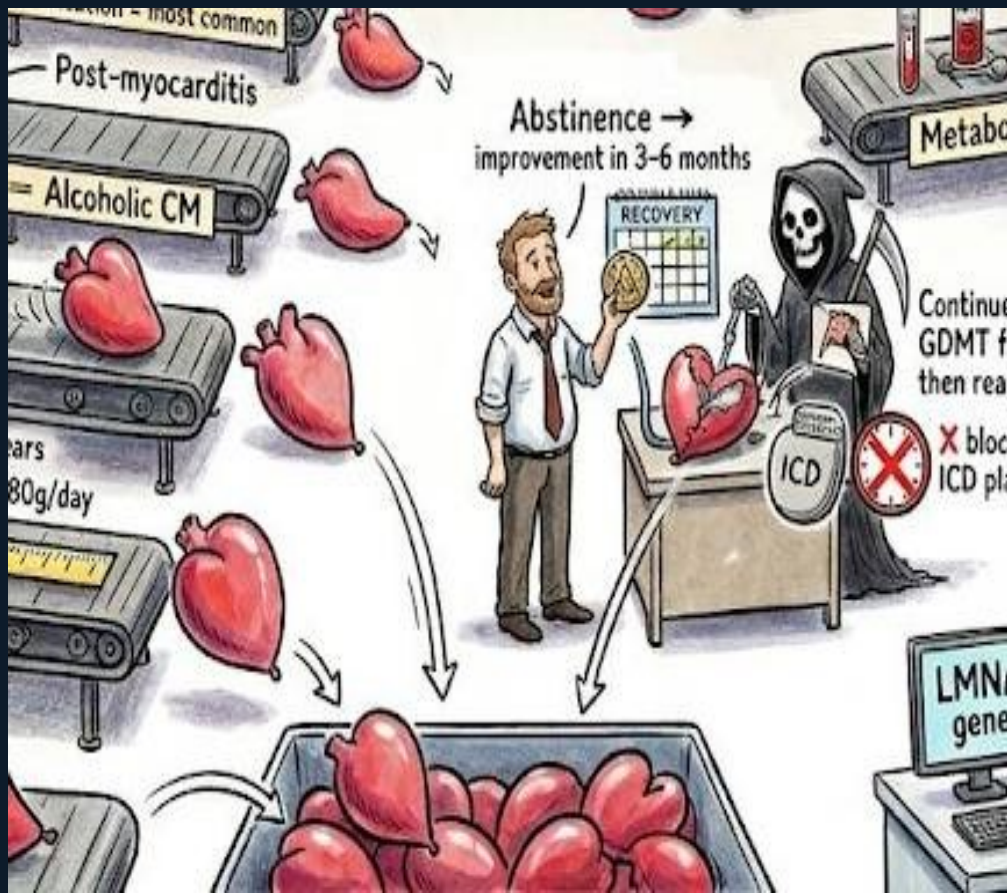




Scene E — DCM: Causes & Alcoholic CM | Industrial Factory

## 9 Belts: The Causes of DCM

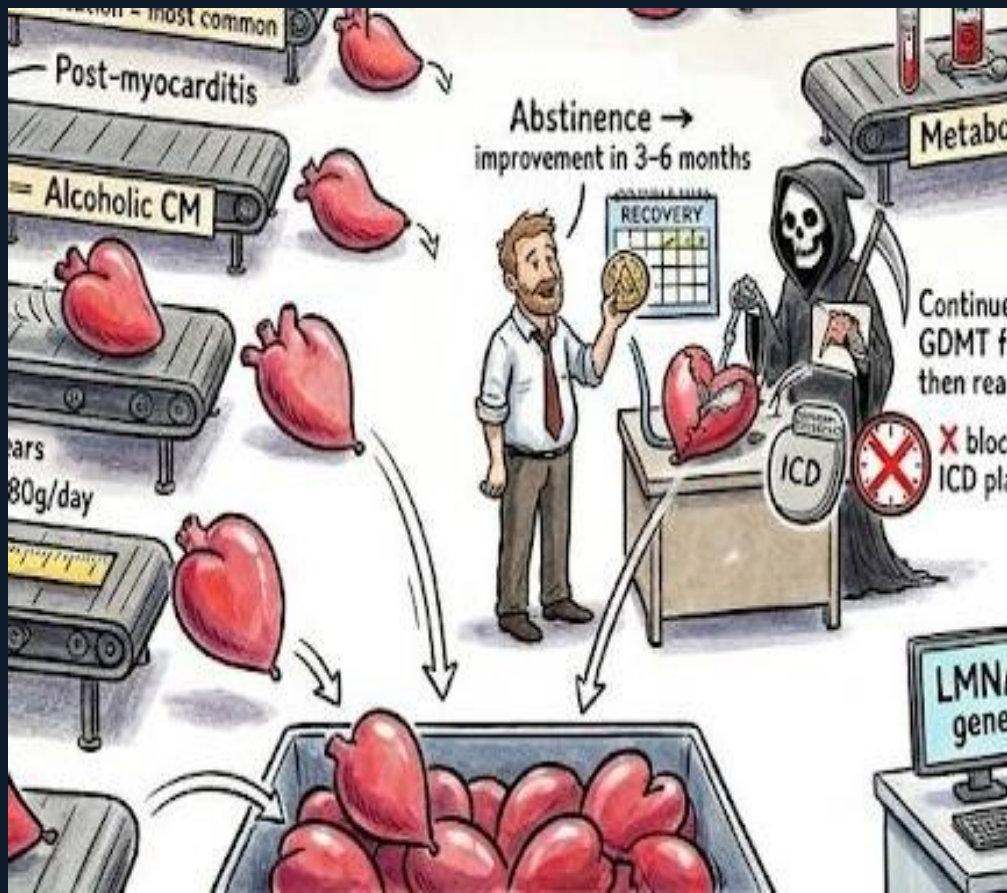
Nine conveyor belts each deliver a different floppy heart to the bin. Belt 1 (blacksmith + blocked pipe): Ischemic — the most common cause; always rule out first with coronary imaging. Belt 2 (DNA helix): Genetic — TTN (titin) truncating variants are the most common genetic cause of DCM, found in ~15–25% of familial and ~3% of sporadic cases. Belt 3 (virus): Post-myocarditis. Belt 4 (whisky barrel): Alcoholic CM — classic threshold  $\geq 60$ –80g/day for  $\geq 5$  years. Belt 5 (Santa Anthracycline rubies): Chemotherapy — doxorubicin-induced cardiomyopathy. The remaining four: tachycardia-induced, peripartum, metabolic/endocrine, HIV/Chagas.



Scene E — DCM: Causes & Alcoholic CM | Industrial Factory

## Alcoholic CM: Recovery Zone

The alcoholic CM character steps off belt 4 into the recovery zone holding a sobriety token: 'Abstinence → improvement in 3-6 months.' A half-repaired floppy heart is being reinflated. This encodes a critical reversibility fact: alcoholic cardiomyopathy can recover substantially with abstinence — unlike most other cardiomyopathies. The grim reaper with a cracked heart represents the opposite trajectory: continued heavy alcohol use = progressive deterioration and poor prognosis. The ICD device with a red X blocking premature placement encodes: trial GDMT for at least 3-6 months with abstinence before committing to ICD — EF may normalize.

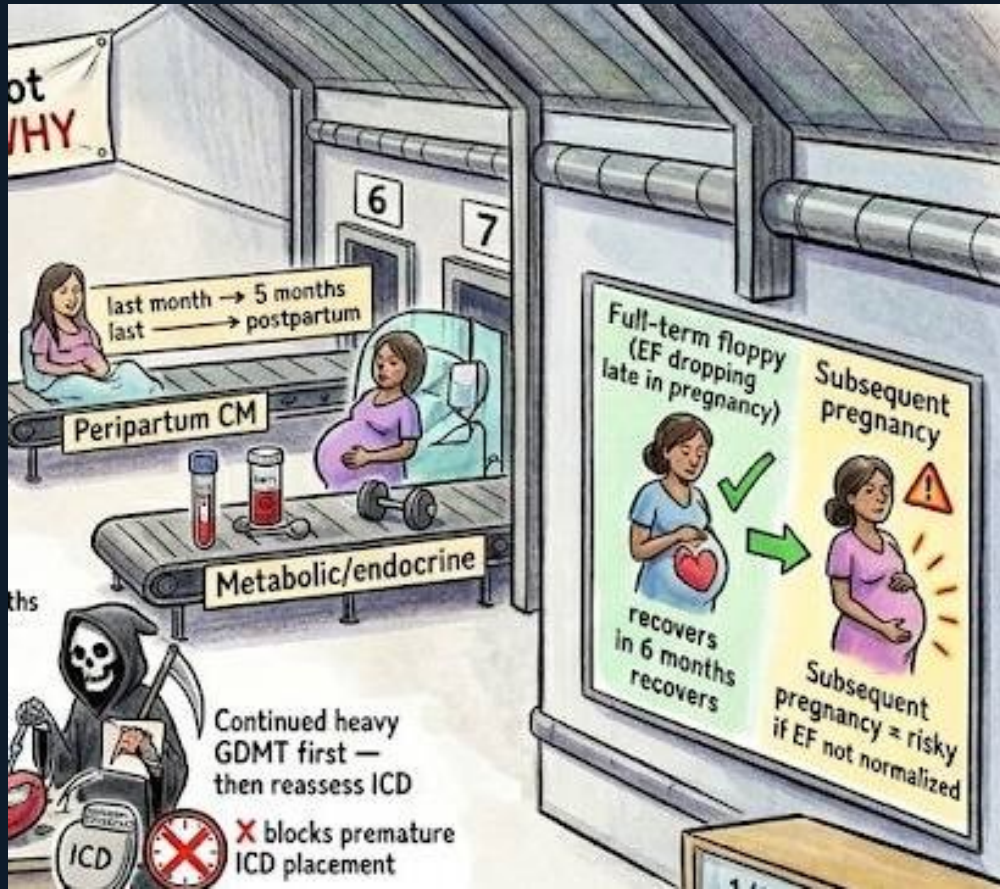


Scene E — DCM: Causes & Alcoholic CM | Industrial Factory

## ICD Timing: GDMT First

The ICD clock labeled ' $\geq 5$  years GDMT first — then reassess ICD' with a red X blocking premature ICD placement encodes one of the most commonly tested board traps in DCM. In newly diagnosed DCM — especially alcoholic, peripartum, or tachycardia-induced — the EF may improve substantially with 3–6 months of guideline-directed medical therapy (GDMT) and trigger removal. Placing an ICD immediately based on a low EF at diagnosis risks unnecessary implantation. Board rule: wait  $\geq 3$  months of optimized GDMT before reassessing ICD eligibility in new DCM — unless the patient has a specific high-risk indication (sustained VT, hemodynamic compromise, or high-risk genetic subtype like LMNA).

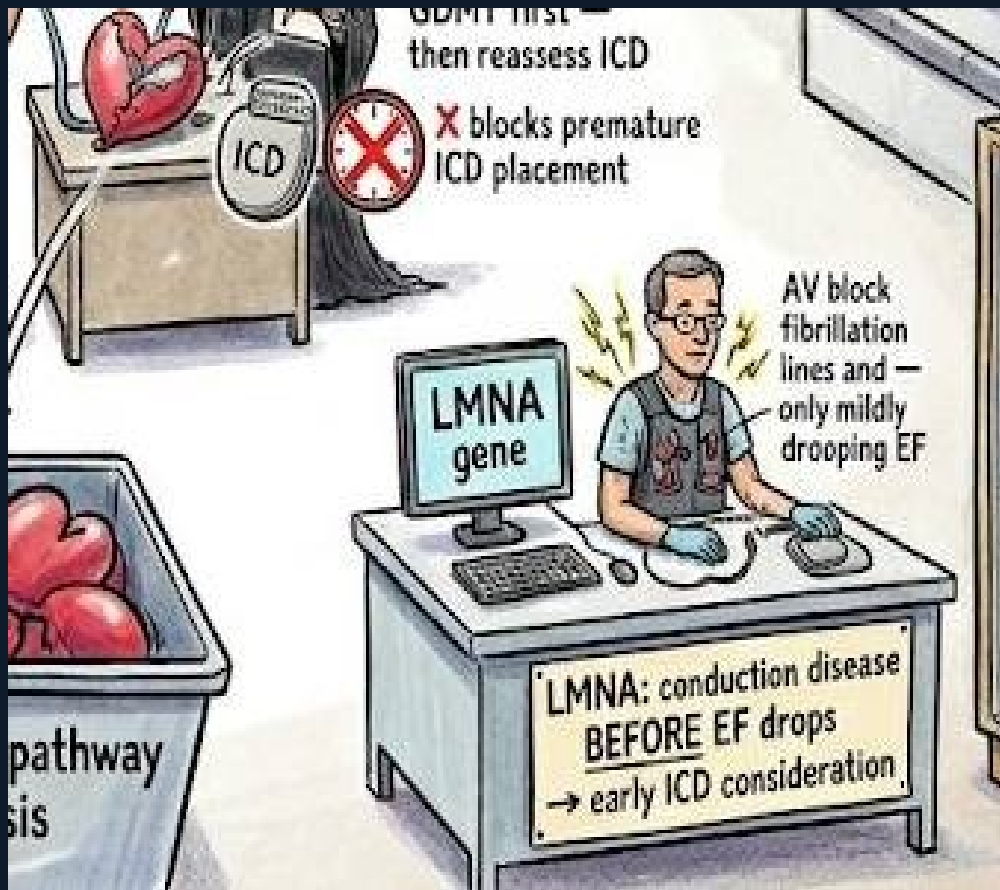




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## Peripartum CM: Timeline + Subsequent Pregnancy

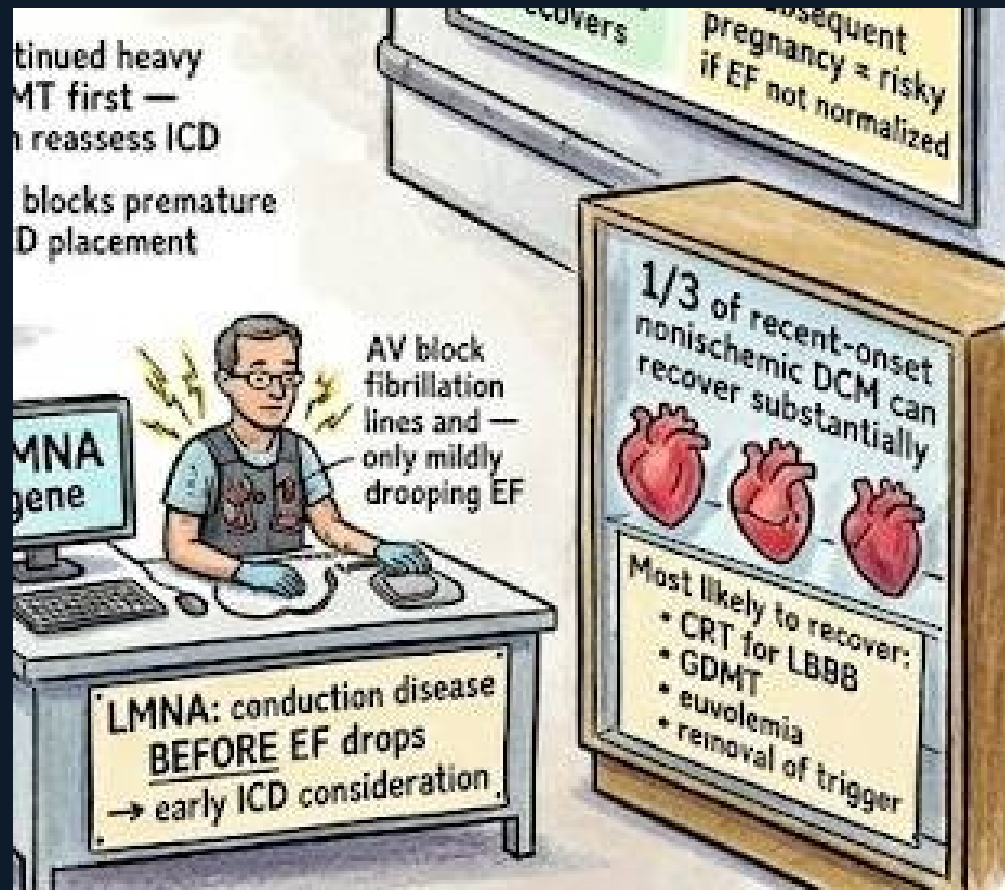
The right wall shows two peripartum CM scenarios. The left figure: EF drops in the last month of pregnancy (full-term floppy heart) and continues through 5 months postpartum — 'last month → 5 months postpartum' is the diagnostic time window. Most recover in 6 months with GDMT and bromocriptine (in some protocols). The right figure: the same patient attempts subsequent pregnancy — a warning triangle appears labeled 'Subsequent pregnancy = risky if EF not normalized.' If EF has fully recovered, subsequent pregnancy is possible but carries risk. If EF remains reduced, pregnancy carries a 25–50% risk of further deterioration and maternal mortality. Never approve subsequent pregnancy without verified full EF normalization.



Scene E — DCM: Causes & Alcoholic CM | Industrial Factory

## LMNA: Conduction Disease BEFORE EF Drops

The LMNA genetics desk shows: a patient with AV block sparking at his chest, atrial fibrillation lines on his shirt, 'only mildly drooping EF' — but an ICD vest is already fitted. Label: 'LMNA: conduction disease BEFORE EF drops → early ICD consideration.' LMNA (lamin A/C) cardiomyopathy follows a distinctive sequence: first AV conduction disease (first-degree → complete AV block), then supraventricular arrhythmias (AF), then ventricular arrhythmias, and only later LV systolic dysfunction. The classic board vignette: middle-aged patient + AV block + family history of SCD + mild LV dysfunction → LMNA. ICD consideration arises earlier than the usual EF  $\leq 35\%$  threshold because arrhythmic SCD risk precedes severe LV dysfunction.



## 1/3 of Recent-Onset Nonischemic DCM Recover

The recovery trophy case on the right wall: three hearts — one fully recovered (green), one partially recovered (yellow), one not recovered (red). Label: '1/3 of recent-onset nonischemic DCM can recover substantially.' Recovery list: CRT for LBBB (cardiac resynchronization therapy dramatically improves EF in LBBB-associated DCM), GDMT, euvolemia, removal of trigger (abstinence from alcohol, stopping tachycardia, treating hypothyroidism, removing the offending chemotherapy). The board tests this as: 'A patient is newly diagnosed with nonischemic DCM — what is the prognosis?' One-third can recover substantially with optimal therapy — prognosis is not uniformly poor.



## Scene E — Quick Reference

### **DCM = phenotype, NOT diagnosis**

Always ask WHY — etiologic workup mandatory

### **TTN truncation = most common genetic**

Titin gene — 15-25% familial, 3% sporadic

### **GDMT first — reassess ICD after $\geq 3$ months**

EF may normalize — avoid premature ICD

### **Subsequent pregnancy → risky if EF not normal**

Verify full EF normalization first

### **1/3 nonischemic DCM recover substantially**

CRT for LBBB, GDMT, euvoemia, trigger removal

### **Ischemic = most common → rule out first**

Coronary imaging before calling it nonischemic DCM

### **Alcoholic CM: $\geq 60$ -80g/day $\times \geq 5$ years**

Abstinence → improvement in 3-6 months

### **Peripartum: last month → 5 months postpartum**

Most recover in 6 months

### **LMNA: conduction disease BEFORE EF drops**

AV block + AF + mild  $\downarrow$  EF → early ICD consideration

### **Continued heavy drinking = poor prognosis**

Grim reaper — abstinence is the intervention